

✓ SOP #5 – Troubleshooting Techniques for Cooling Towers

Applies to: Induced-draft, forced-draft, crossflow, counterflow, hybrid and fluid cooler towers

Trigger Events: High condenser water temps, excess drift, vibration, poor heat rejection, or microbial/algae growth

🔧 1. Common Failure Signs & Diagnoses

Symptom	Possible Cause	Recommended Action
High condenser return temp	Dirty fill media, low airflow	Clean media, check fan speed & direction
Visible drift or water loss	Damaged eliminators, overflow, wind blow-off	Replace drift eliminators, adjust level
Excess vibration or noise	Fan imbalance, worn motor bearings	Rebalance, grease, or replace motor
Overflow or low basin level	Faulty makeup valve, float stuck	Adjust or replace makeup float assembly
Biological growth or smell	Inadequate water treatment	Test water, shock dose biocide